

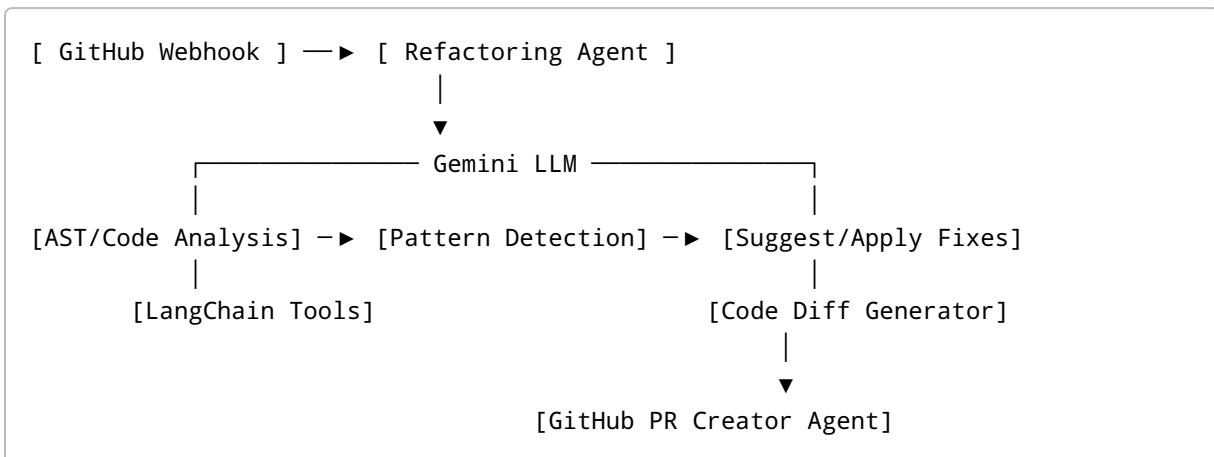
Project Documentation: Autonomous Code Refactoring Agent

Objective

Build an **Autonomous AI Agent** that:

- Monitors a GitHub repository for code updates.
- Analyzes the code using Gemini and AST/CodeBERT.
- Detects code smells, outdated patterns, or inefficiencies.
- Automatically refactors the code.
- Submits a Pull Request with detailed explanations.

System Architecture



Tech Stack

Component	Tool/Framework
Agent Framework	LangChain + LangGraph
LLM	Gemini Pro via API
Code Analysis	<code>ast</code> module / CodeBERT
Git Integration	GitHub API
CI/CD Hook	GitHub Webhook / Actions

Component	Tool/Framework
Memory	ChromaDB or local
Hosting	FastAPI + Docker

Module Breakdown

1. Repo Watcher (Event Trigger)

- **Trigger:** GitHub Webhook (push/PR events)
- **Action:** Sends file changes to the refactoring pipeline

2. Code Analyzer Agent

- **Input:** Raw source code or code diff
- **Tools:**
 - Python AST for static analysis
 - Gemini LLM for semantic analysis
 - CodeBERT (optional) for deep context

3. Refactoring Agent

- Uses Gemini to:
 - Detect anti-patterns
 - Apply refactoring suggestions
 - Annotate code with meaningful comments

4. Pull Request Agent

- Compares original vs. refactored code
- Commits changes to a new branch
- Submits PR via GitHub API
- Adds summary of changes and reasoning

5. CI/CD Integration (Optional)

- GitHub Actions to test refactored code
 - Notifies via Slack or merges if successful
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Sample Prompt (Gemini)

You are an expert Python developer. Analyze the following code for anti-patterns or inefficiencies.

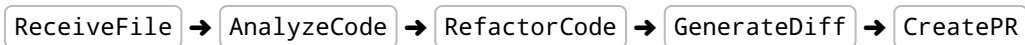
Respond with:

1. Issues found
2. Refactored version
3. Explanation for each change

Code:

<insert_code_here>

Agent Workflow (LangGraph)



Development Phases

Phase	Description	Output
1	GitHub repo listener & file fetcher	Trigger-ready FastAPI
2	AST + Gemini-based code analyzer	Detected smells/issues
3	Refactoring via Gemini + LangChain	Cleaned code output
4	GitHub PR agent	Auto-submitted PRs
5	Full pipeline test on real project	Working agentic loop
6	Optional: CI/CD + Slack alerts, metrics	Fully autonomous dev loop

Next Step

Proceed with Phase 1: Set up FastAPI service and GitHub webhook handler.

Let me know if you want full repo scaffolding or sample code for any phase.